Content

Back Matter **3**

Back Matter **4**

23 Extensions and Open Problems **4**

23.1 The Dynamic Market Model: When Φ Changes Over Time 4

23.2 Multi-Language Resolution: When the Substrate Shifts 5

23.3 AI-Generated Names and the Semantic Entropy Floor 5

23.4 Voice Search and the Phonetic Resolution Problem 6

23.5 Resolution in Non-English Registry Systems 6

23.6 Open Problems for Future Research 7

Conclusion **8**

A Master Equation Reference Card **11**

B Complete Symbol Glossary **14**

C Industry Friction Coefficients Φ by Sector **18**

D The Resolution Score Quick-Reference Calculator **20**

E Google Business Category Index (Selected High-Traffic Categories) **22**

F Failure Mode Diagnostic Flowchart **25**

G Suggested Courses and Curriculum Maps **27**

Bibliography **30**

Back Matter

CHAPTER 23

Extensio and Open Problems

Chapter Purpose

The framework in this textbook is complete in its current scope. But commercial naming is not a static domain. Classification systems evolve, search architectures change, new media environments create new resolution requirements, and the global expansion of AI-mediated search introduces challenges that no existing framework fully addresses. This chapter identifies the most important extensions and open problems at the boundary of the current theory.

23.1 The Dynamic Market Model: When Φ Changes Over Time

Throughout this textbook, sector friction Φ has been treated as a quasi-static variable: measurable at a point in time, subject to slow change. In practice, Φ can change rapidly due to:

- A dominant national brand entering a previously thin local category (sudden Φ increase)
- Regulatory changes that restructure competitive density (variable depending on regulation type)

- Platform algorithm updates that shift which entities receive classification priority
- Category emergence: a new service category splits off from an existing one, creating two thinner markets from one denser one (sudden Φ decrease in the new category)

The open problem is the formalization of $\Phi(t)$ as a time-varying function and the derivation of the conditions under which a Stable Atom entity in a low-friction market retains its position as that market's friction rises. Preliminary analysis suggests a **Classification Momentum** term $\kappa \cdot \alpha \cdot t_{established}$ that represents the accumulated classification confidence an established entity has built, which resists displacement even as Φ rises around it.

23.2 Multi-Language Resolution: When the Substrate Shifts

The Minimal Resolution Template $V^* = C_l + O_s + F_a + N_c$ was derived from the structure of the resolution fields as they operate in English-language commercial environments. The fundamental field structure — search, registry, task, intent — is language-independent. But the token-to-vector mappings, the cluster positions of nouns and verbs, and the basin structure of the embedding space all vary by language.

The open problem is whether the four irreducible components hold across languages, and what the cross-language drift characteristics are for businesses that operate in bilingual markets. Preliminary evidence from Spanish-English bilingual markets suggests that the resolution hierarchy changes: geographic context (C_l) becomes even more important in bilingual environments because it provides disambiguation that category nouns alone cannot achieve across language boundaries.

23.3 AI-Generated Names and the Semantic Entropy Floor

Large language model-based naming tools, which are becoming increasingly common, tend to generate names with a characteristic failure mode: they optimize for human aesthetic preferences (phonetic pleasantness, novelty, memorability) while producing systematically low α scores. The reason is structural: the training data for creative naming tools does not include resolution field performance data, so the models have no signal to optimize against.

The open problem is whether resolution field performance scores can be incorporated into the reward function for naming generation models, producing AI tools that generate names satisfying both aesthetic preferences and resolution requirements simultaneously. This is the generative inverse of the scoring tool developed in this textbook.

23.4 Voice Search and the Phonetic Resolution Problem

Voice search queries follow a different structural pattern than text queries. They are more conversational, longer, and more frequently include complete sentences ("find me a plumber near me who can come today"). The action verb (F_a) component is more prominent in voice queries because spoken requests naturally take verb-first or verb-anchored structures.

The open problem is the derivation of a voice-query-weighted version of the alignment coefficient α_{voice} that weights the F_a component more heavily and the N_c component slightly less heavily, reflecting the structural differences between spoken and typed queries. As voice search continues to grow as a primary search modality, businesses in high-R_p categories (where the urgency-driven customer may be speaking into their phone while standing in front of the broken pipe) will face increasingly specific resolution requirements.

23.5 Resolution in Non-English Registry Systems

NAICS, GBP, and O*NET are primarily English-language systems. The European equivalent (NACE codes), the UN's ISIC classification, and regional registry systems each have their own token-to-category mappings. The Minimal Resolution Template applies in principle, but the specific token vocabulary for N_c and F_a changes by classification system.

The most important open problem in this domain is the mapping of resolution component tokens across multiple international registry systems simultaneously, for businesses that operate across language and regulatory boundaries.

23.6 Open Problems for Future Research

The following are formally open problems within the Thermodynamics of Commercial Identity framework:

1. **The Classification Momentum Theorem**: Formalize the conditions under which an established Stable Atom entity retains its position as sector friction Φ rises, and derive the decay function for accumulated classification confidence.

2. **The Multi-Language Closure Proof**: Determine whether the four-component closure set $\{N_c, O_s, F_a, C_l\}$ is invariant across language families, or whether different languages require different irreducible component sets.

3. **The Generative Inverse**: Derive the exact specification for a name generation model that optimizes simultaneously for alignment (α), drift (σ), and transmissibility (T) within a given sector friction environment.

4. **The Acquired Alignment Rate ρ**: Empirically measure the market learning rate ρ across industry categories, providing a data-grounded calibration for the acquired alignment equation $\alpha_t = \alpha_0 + \rho m$.

5. **The Portfolio Interference Matrix**: Formalize the conditions under which two entities in the same portfolio compete for the same classification basin, and derive the optimal naming distance required to prevent basin overlap.

6. **The Voice-Query Alignment Correction**: Derive α_{voice} as a function of α_{text} and the structural differences between spoken and typed query patterns.

Conclus

The Deterministic Principle and What It Means

What This Framework Changes About How You Name Things

Before this framework, naming a business was a decision made by feel. You gathered opinions. You checked domain availability. You asked friends. You hired a consultant who asked about your brand values and your target customer's lifestyle aspirations. The result was a name that people liked, or that sounded good, or that felt right — and no one could tell you, with any precision, whether it would work.

After this framework, naming a business is an engineering decision with measurable inputs, derivable outputs, and falsifiable predictions. You compute the alignment score. You check the drift. You calculate the trough depth. You run the atmospheric resultant. You score the viability. The result is a number that predicts specific commercial outcomes: the rate of organic search visibility, the cost multiplier for customer acquisition, the time to market legibility, and the annual overhead the naming decision will impose on the business's marketing budget.

The engineering approach does not eliminate judgment. It does not replace the creative instinct for what sounds right in the mouth or sits well on a card. What it eliminates is the blind spot: the possibility of choosing a name that sounds right and works against you commercially from the day you register it.

The Full Law, Stated Simply

Everything in this textbook reduces to one structural claim:

A business name is a thermodynamic input into a multi-substrate resolution environment. The environment evaluates it deterministically. The outcome of that evaluation — classification accuracy, search visibility, conversion probability, persistence over time — is a function of measurable structural properties of the name string, not of its aesthetic quality.

The full law:

$$\text{Success} \propto \text{Pressure} \times \text{Resolution Fit} \times \text{Execution} - \text{Resistance}$$

Or in the complete mathematical form:

$$S_x(t) = \int_0^t \frac{(E + \Delta A_r R_p)(\alpha(1 + \omega R_p) + \lambda \Delta\, \chi(R_p))}{\Phi(1 + \sigma(1 + \psi R_p))^2}\, dt - R_d$$

The optimal name that minimizes resistance from the start:

$$V^* = C_l + O_s + F_a + N_c$$

Naming as the First Engineering Decision of a Business

The naming decision is made before the website is built, before the first customer is acquired, before the first marketing dollar is spent. It is the only decision that is simultaneously permanent and foundational: permanent because renaming an established entity is costly and slow; foundational because every subsequent marketing investment either compounds on it (if the name is correct) or compensates for it (if it is not).

This is why naming deserves the same rigor applied to any other foundational business decision. A founding team that spends months analyzing market size, competitive positioning, pricing strategy, and go-to-market approach — and then names the company in an afternoon based on what sounds cool — has applied first-principles thinking to every variable in the success equation except the one that multiplies all the others.

Final Statement: When Name ≡ Category ≡ Object ≡ Function

When the name achieves Algorithmic Autonomy — when the entity's name is so precisely aligned with the resolution environment that the system maintains it, classifies it, and surfaces it without ongoing human intervention — the company enters a different operating mode.

Marketing effort stops compensating and starts compounding. Every dollar of advertising builds on a foundation that the name has already established, rather than rebuilding the foundation each month. Every new review confirms a classification the algorithm already holds. Every referral repeats a message the name already teaches.

This is not a theoretical ideal. It is an achievable engineering outcome. It is achieved, systematically and repeatably, by applying the Minimal Resolution Template to every naming decision, auditing every existing name against the viability scoring criteria, and treating every point of alignment deficit as a measurable, addressable, solvable structural problem.

When Name ≡ Category ≡ Object ≡ Function:

The system no longer needs to infer.
The market no longer hesitates.
The entity becomes the default answer.

That is Algorithmic Autonomy.
That is the goal.
That is what naming, done correctly, achieves.

Armstrong Knight
intent-tensor-theory.com

CHAPTER A

Master Equation Reference Card

A single-page reference containing every major equation in the textbook, organized by function. Intended for use alongside the scoring worksheet and audit protocol.

Resolution Structure

$$\mathcal{F} = \mathcal{F}_{search} \cup \mathcal{F}_{registry} \cup \mathcal{F}_{task} \cup \mathcal{F}_{intent}$$

$$V^* = C_l + O_s + F_a + N_c \quad \text{(Minimal Resolution Template)}$$

$$\text{Entity Resolution} = \bigcap_i \mathcal{F}_i(N)$$

Core Coefficients

$$\alpha = \frac{1}{|\mathcal{F}|} \sum_i P_i(N) \quad \text{(semantic alignment)}$$

$$\sigma = \|\vec{v}_{name} - \vec{v}_{intended}\| \quad \text{(vector drift)}$$

$$\Phi \in [0, 1] \quad \text{(sector friction, from Chapter 8 table)}$$

Cost Equations

$$D_t = \Phi(1 - \alpha) \quad \text{(trough depth)}$$

$$W_\Delta = \frac{1}{\alpha^2} \quad \text{(work-offset ratio)}$$

$$\tau_i^{monthly} = E_0\left(\frac{1}{\alpha^2} - 1\right) \quad \text{(monthly inference tax)}$$

$$R_{sn} = \frac{\alpha - \sigma}{\Phi(1 + \sigma)} \quad \text{(atmospheric resultant)}$$

$$R_d = R_{base}(1 + \gamma(1 - \alpha)) \quad \text{(decay rate)}$$

Differentiation

$$\Delta = U \cdot M \cdot T, \quad U = 1 - O_c, \quad M = \frac{R_1 + R_7 + R_{30}}{3}, \quad T = \frac{V_s}{F_c}$$

$$\lambda = C_r \cdot C_m, \quad P_\Delta = \frac{\Delta}{\Phi}$$

$$\Delta_{required} = \frac{\Phi(1 + \sigma)^2 - \alpha}{\lambda} \quad \text{(compensation requirement)}$$

$$\Delta_{eff} = \frac{\Delta}{1 + k\Delta} \quad \text{(saturation correction)}$$

Viability Scoring

$$V = \alpha + \lambda\Delta - \mu(1 - \alpha)\sigma, \quad \mu = 0.25$$

$$B_e = \frac{\Phi(1 + \sigma)^2}{\alpha + \lambda\Delta} \quad \text{(execution burden)}$$

$$D'_t = \max(0, \Phi(1 - (\alpha + \lambda\Delta))) \quad \text{(adjusted trough)}$$

$$C_h = L \cdot \frac{\alpha + \lambda\Delta}{\Phi(1 + \sigma)} \quad \text{(crest height)}$$

Resolution Pressure

$$R_p = N \cdot U_r \cdot T_c, \quad I_s = \max(0, R_p - \eta), \quad V_s = \frac{R_p}{F_{cog}}$$

$$\alpha_p = \alpha(1 + \omega R_p), \quad \sigma_p = \sigma(1 + \psi R_p), \quad \chi(R_p) = 1 + \nu R_p - \zeta R_p^2$$

The Master Equation

$$S_x(t) = \int_0^t \frac{(E + \Delta A_r R_p)(\alpha(1 + \omega R_p) + \lambda \Delta\, \chi(R_p))}{\Phi(1 + \sigma(1 + \psi R_p))^2}\, dt - R_d$$

Acquired Alignment and Persistence

$$\alpha_t = \alpha_0 + \rho m \quad \text{(acquired associative alignment)}$$

$$P_s(t) = \int_0^t \frac{E \cdot \alpha}{\Phi(1 + \sigma)^2}\, dt - R_d \quad \text{(state persistence, simplified)}$$

Viability Bands Quick Reference

$V \geq 0.85$	Stable Atom	Algorithmic Autonomy
$0.65 \leq V < 0.85$	Transitional	Moderate ongoing effort
$0.45 \leq V < 0.65$	High Effort	Significant structural overhead
$V < 0.45$	Ghost Risk	Rename strongly indicated

CHAPTER B

Complet

Symbol Glossary

Symbol	Name	Definition and Range
V^*	Min. Resolution Template	Optimal resolved name: $C_l + O_s + F_a + N_c$
N_c	Category Noun	Maps to industry identity in registry systems
O_s	Object	Transaction anchor; the economic unit acted upon
F_a	Function	Operation performed; task field signal
C_l	Context	Geographic or audience scope; search-space reduction
α	Alignment	Match between name and field expectations; $[0, 1]$
α_0	Intrinsic Alignment	Alignment from name content alone at launch
α_t	Acquired Alignment	Effective alignment after m messaging cycles
α_p	Pressure-Adjusted Alignment	$\alpha(1 + \omega R_p)$; alignment amplified by urgency
σ	Drift	Displacement of name vector from intended category; $[0, 1]$

Symbol	Name	Definition and Range
σ_p	Pressure-Weighted Drift	$\sigma(1+\psi R_p)$; drift amplified by urgency
Φ	Sector Friction	Competitive resistance of market environment; $[0,1]$
τ_i	Inference Tax	Monthly cost of naming ambiguity in effort/spend
D_t	Trough Depth	Visibility deficit at launch: $\Phi(1-\alpha)$
D'_t	Adjusted Trough	Trough with differentiation: $\max(0, \Phi(1-\alpha-\lambda\Delta))$
W_Δ	Work-Offset Ratio	Effort multiplier: $1/\alpha^2$
R_{sn}	Atmospheric Resultant	Net resolution force: $(\alpha-\sigma)/[\Phi(1+\sigma)]$
R_d	Decay Rate	Erosion rate of unreinforced market signals
E	Execution Effort	Direct marketing and distribution force applied
E_{eff}	Effective Effort	$E+\Delta A_r R_p$; includes narrative lift
V	Viability Score	$\alpha+\lambda\Delta-\mu(1-\alpha)\sigma$
Δ	Differentiation Force	$U \cdot M \cdot T$; compensatory market power
Δ_{eff}	Effective Differentiation	$\Delta/(1+k\Delta)$; saturation-corrected
Δ_p	Pressure-Convertible Δ	$\Delta \cdot \chi(R_p)$; differentiation under pressure
U	Uniqueness	$1-O_c$; non-substitutability of positioning
M	Memorability	$(R_1+R_7+R_{30})/3$; average recall probability
T	Transmissibility	V_s/F_c; ease of accurate restatement
λ	Conversion Constant	$C_r \cdot C_m$; fraction of Δ that converts to traction
C_r	Category Receptivity	Degree to which market rewards this differentiation type
C_m	Medium Compatibility	Appropriateness of communication channel
P_Δ	Differentiation Pressure	Δ/Φ; differentiation vs. friction diagnostic

Symbol	Name	Definition and Range
μ	Ambiguity Burden	Penalty multiplier for low-α, high-σ combinations; ≈ 0.25
F_m	Market Force	$\alpha + \lambda\Delta - \mu(1-\alpha)\sigma$
B_e	Execution Burden	$\Phi(1+\sigma)^2/(\alpha+\lambda\Delta)$; effort efficiency diagnostic
N_l	Narrative Lift	$\Delta \cdot A_r$; organic distribution from resonant differentiation
A_r	Audience Resonance	Degree to which differentiation activates target market
k	Saturation Constant	Governs diminishing returns of differentiation
L	Market Ceiling	Maximum attainable share in reachable field
C_h	Crest Height	$L\cdot(\alpha+\lambda\Delta)/[\Phi(1+\sigma)]$; maximum market position
$S_x(t)$	Market Success	Accumulated market position function over time
$P_s(t)$	State Persistence	Probability of maintained market recognition at time t
R_p	Resolution Pressure	$N \cdot U_r \cdot T_c$; market activation force
N	Necessity Intensity	How essential the problem is; $[0,1]$
U_r	Urgency	Speed with which resolution is required; $[0,1]$
T_c	Consequence	Cost of delay; $[0,1]$
η	Activation Threshold	Minimum R_p for search behavior to emerge
I_s	Search Intent Emergence	$\max(0, R_p - \eta)$; activated market intensity
V_s	Searcher Velocity	R_p/F_{cog}; speed of buyer selection
F_{cog}	Cognitive Friction	Mental effort required to parse the name
$\chi(R_p)$	Pressure Conversion	$1 + \nu R_p - \zeta R_p^2$; differentiation-pressure function
ω	Alignment Sensitivity	Rate of alignment amplification under pressure
ψ	Drift Amplification	Rate of drift cost amplification under pressure
ν	Pressure Gain	Positive gain coefficient for utility differentiation under pressure

Symbol	Name	Definition and Range
ζ	Collapse Factor	Ornamental differentiation collapse coefficient under pressure
ρ	Learning Rate	Rate of acquired associative alignment per messaging cycle
m	Messaging Cycles	Cumulative major exposure cycles completed
O_c	Overlap Coefficient	Degree of positioning overlap with competitors
R_1, R_7, R_{30}	Recall Probabilities	Retention probability at 1, 7, 30 days respectively
B_c	Selection Bias	$R_p(1-\sigma)$; urgency-driven preference for clarity
β	Recognition Baseline	Minimum cycles for any entity to achieve classification
T_u/T_r	Time to Understanding	Cycles to market legibility
$\mathcal{F}$	Resolution Field Union	$\mathcal{F}_{search} \cup \mathcal{F}_{registry} \cup \mathcal{F}_{task} \cup \mathcal{F}_{intent}$
$\vec{v}_i$	Token Vector	Embedding of word i in $\mathbb{R}^n$
ϵ_{min}	Tear Threshold	Minimum projection amplitude for field classification

CHAPTER C

Industry Friction Coefficients Φ by Sector

These estimated Φ ranges reflect average competitive density, category maturity, and query saturation. Apply the sector friction rubric from Chapter 18 to adjust within the range based on specific market characteristics. Use the midpoint as a default starting value.

Industry / Service Category	Φ Low	Φ High	Friction Level
Personal injury law	0.93	0.97	Very High
Search engine optimization	0.91	0.95	Very High
Financial planning / wealth management	0.87	0.92	Very High
Digital marketing agencies	0.83	0.89	High
Business consulting (general)	0.82	0.88	High
Real estate services	0.82	0.90	High
IT services / managed services	0.80	0.87	High
Health coaching / wellness	0.78	0.85	High
Tax preparation (national)	0.76	0.84	High
Staffing and recruiting	0.74	0.82	High
Corporate management training	0.72	0.80	Medium-High

Industry / Service Category	**Φ Low**	**Φ High**	**Friction Level**
Residential roofing (dense metro)	0.68	0.76	Medium-High
Plumbing services	0.65	0.73	Medium
HVAC services	0.63	0.71	Medium
Electrical contractors	0.62	0.70	Medium
Landscaping services	0.60	0.68	Medium
Commercial cleaning	0.58	0.66	Medium
Auto body repair	0.57	0.65	Medium
Catering services	0.55	0.63	Medium
Tax preparation (local, seasonal)	0.54	0.62	Medium
Pet grooming	0.52	0.62	Medium-Low
Specialty food retail	0.48	0.58	Medium-Low
Personal training (local)	0.45	0.55	Medium-Low
Commercial roofing (niche)	0.44	0.54	Medium-Low
Specialty B2B services	0.40	0.55	Medium-Low
Divorce financial planning (niche)	0.38	0.50	Low-Medium
Emerging professional services	0.30	0.50	Low
New digital categories (< 5 yr)	0.20	0.40	Low
Genuinely new category (pioneer)	0.10	0.25	Very Low

Adjustment factors: Subtract 0.08–0.12 from midpoint for rural or small-metro markets. Add 0.05–0.10 for markets with dominant national franchises. Subtract 0.10–0.15 for a genuine niche sub-specialization that eliminates most of the category's competitive density.

CHAPTER D

The Resolution Score Quick-Reference Calculator

A fast-scoring reference for practitioners who need a viability estimate without running the full audit. Input five numbers, output the Viability Score and viability band in under five minutes.

Five-Minute Scoring Protocol

Input 1 — α estimate (alignment):
Ask: *Does the name contain a clear category noun, a function word, and an object? Is it geographically scoped for a local business?*

- All four components clear: $\alpha = 0.90$
- Three components clear: $\alpha = 0.75$
- Two components clear: $\alpha = 0.55$
- One component or less: $\alpha = 0.25$

- Invented word / no components: $\alpha = 0.08$

Input 2 — σ estimate (drift):
Ask: *Do any words in the name have strong associations with a different industry?*

- No competing associations: $\sigma = 0.05$
- One word has mild secondary associations: $\sigma = 0.25$
- One word drifts moderately toward a competing field: $\sigma = 0.50$
- One or more words strongly associated with wrong industry: $\sigma = 0.75$
- Name is entirely a competing category's vocabulary: $\sigma = 0.90$

Input 3 — Φ estimate (friction):
Use midpoint from Appendix C table. Default: $\Phi = 0.70$.

Input 4 — Δ estimate (differentiation):
Ask: *Does the business have a genuinely unique, memorable, one-sentence proposition?*

- Elite proposition, highly transmissible: $\Delta = 0.75$
- Strong proposition, fairly transmissible: $\Delta = 0.55$
- Moderate distinction, somewhat transmissible: $\Delta = 0.35$
- Generic claims ("great service"): $\Delta = 0.05$

Input 5 — λ estimate (conversion):
Ask: *Does your market actively reward this type of distinction?*

- High receptivity, right channel: $\lambda = 0.80$
- Moderate receptivity: $\lambda = 0.55$
- Low receptivity or wrong channel: $\lambda = 0.25$

Compute:

$$V = \alpha + \lambda\Delta - 0.25(1-\alpha)\sigma$$

$$D_t = \Phi(1-\alpha)$$

$$R_{sn} = \frac{\alpha - \sigma}{\Phi(1+\sigma)}$$

Interpret using the viability band table from Chapter 18. If $R_{sn} < 0$: name is net-negative regardless of V score — renaming is indicated.

CHAPTER E

Google Business Category Index (Selected High-Traffic Categories)

The GBP system contains over 4,200 categories. This appendix lists the most commercially significant categories for consumer and small-business services, with the primary token alignment signal for each. The full category list is maintained at intent-tensor-theory.com/registry.

GBP Category String	Primary Token(s)	Secondary Token(s)
Roofing contractor	Roof, Roofing	Repair, Install, Contractor
Plumber	Plumb, Plumbing	Pipe, Drain, Water
Electrician	Electric, Electrical	Wire, Panel, Install
HVAC contractor	HVAC, Heating, Cooling	Air, System, Unit
General contractor	Contractor, Construction	Build, Remodel, Renovation
Painting contractor	Paint, Painting	Interior, Exterior, Coat
Flooring contractor	Floor, Flooring	Install, Hardwood, Tile

GBP Category String	Primary Token(s)	Secondary Token(s)
Landscaping company	Landscape, Landscaping	Lawn, Garden, Maintenance
Cleaning service	Clean, Cleaning	Maid, Janitorial, Sanitize
Pest control service	Pest, Exterminator	Control, Treatment, Inspect
Law firm	Law, Legal, Attorney	Firm, Practice, Lawyer
Divorce lawyer	Divorce, Family Law	Attorney, Legal, Counsel
Personal injury attorney	Personal Injury, Accident	Attorney, Lawyer, Legal
Accountant	Account, Accounting, CPA	Tax, Financial, Bookkeep
Tax preparation service	Tax, Taxes	Preparation, Service, Filing
Financial planner	Financial, Finance	Planning, Advisor, Wealth
Insurance agency	Insurance	Agency, Agent, Coverage
Real estate agency	Real Estate, Realty	Agent, Property, Homes
Property management company	Property Management	Rental, Tenant, Landlord
Management consultant	Consulting, Consultant	Management, Advisory, Business
Marketing agency	Marketing, Advertising	Agency, Digital, Campaign
IT service provider	IT Services, Technology	Computer, Network, Support
Software company	Software, App, Application	Development, Technology
Staffing agency	Staffing, Recruiting	Agency, Employment, Talent
Auto repair shop	Auto, Automotive, Car	Repair, Mechanic, Service
Auto body shop	Auto Body, Collision	Repair, Paint, Restore
Dentist	Dental, Dentist	Tooth, Teeth, Oral
Chiropractor	Chiropractic, Chiropractor	Spine, Adjustment, Therapy
Physical therapist	Physical Therapy	Rehabilitation, Therapy, Exercise
Gym	Gym, Fitness	Exercise, Workout, Training

GBP Category String	Primary Token(s)	Secondary Token(s)
Personal trainer	Personal Training, Trainer	Fitness, Workout, Coach
Hair salon	Hair, Salon	Stylist, Cut, Color
Barber shop	Barber, Barbershop	Hair, Cut, Shave
Restaurant	Restaurant	Dining, Food, Cuisine
Caterer	Catering, Caterer	Event, Food, Service
Bakery	Bakery, Bakeries	Bread, Cake, Pastry
Pet grooming service	Pet Grooming, Groomer	Dog, Cat, Wash, Trim
Veterinarian	Vet, Veterinary, Animal	Clinic, Hospital, Care
Trucking company	Trucking, Freight	Transport, Haul, Delivery
Moving company	Moving, Movers	Relocation, Pack, Transport

CHAPTER F

Failure Mode Diagnostic Flowchart

A textual representation of the routing protocol for identifying the primary failure mode of any named entity. Follow each branch to its terminal failure mode identification.

START: Does the business appear on first-page results for its category + city search?
YES $\rightarrow$ Naming is not the primary growth constraint. Check for FM5 (Decorative Differentiation), FM6 (Non-Convertible Differentiation), or FM10 (Acquired Debt if performance is declining). Run viability score to confirm $V \geq 0.65$.

NO / SOMETIMES $\rightarrow$ Proceed to Q2.

Q2: Read the business name aloud to a stranger with no additional context. What is their response?
They correctly name the industry $\rightarrow$ Category noun is present (N_c exists). Proceed to Q3 to identify which other component is missing.

They name a different industry $\rightarrow$ Unintended Basin (FM9) or Ambiguity Overload (FM7). Run basin test. If $\sigma > 0.55$, calculate ambiguity penalty. If penalty $> \lambda\Delta$: FM7 confirmed, rename required.

No confident guess → Dimensional Tear (FM8) or Missing Category (FM1). Proceed to Q4.

Q3: A stranger can identify the industry but cannot identify the specific service. Which question can they NOT answer?
Cannot identify what action is performed → FM2: Missing Function ($F_a = \emptyset$). Add function word.

Cannot identify what is being acted upon → FM3: Missing Object (O_s absent or generic). Specify the object.

Cannot identify who the service is for or where → FM4: Missing Context ($C_l = \emptyset$). Add geographic or audience context if serving a local or specific market.

Q4: Enter the business name in GBP category auto-suggest. What is returned?
Correct category on first suggestion → Not a Tear. Return to Q3 — one component is missing but the category noun is functioning.

Wrong but specific category → FM9: Unintended Basin. The name has been captured by a competing category. Remove or replace the drifting token.

"Business" or no suggestion → FM8: Dimensional Tear or **FM1: Missing Category**. Run three-test diagnostic from Chapter 15 to confirm.

- If three-test fails completely: FM8 Tear confirmed. Rename.
- If three-test passes partially: FM1 Missing N_c. Add category noun.

SECONDARY CHECK: Is the business's organic search performance declining despite no change in marketing?
YES → FM10: Acquired Debt. The name's effective σ has increased due to environmental change. Audit current basin alignment against today's competitive landscape and update the name if needed.

NO → Failure mode identified in primary routing above. Proceed to the repair protocol for that failure mode.

CHAPTER G

Suggest

Courses and Curriculum Maps

Three curriculum structures for using this textbook in academic and professional education settings: a four-week accelerated module, a full-semester course, and a graduate-level seminar.

Structure A: Four-Week Accelerated Module (Professional Development)

Week 1 — The Resolution Framework Sessions: Foreword, Chapters 1–4. Objective: Understand the multi-field resolution environment, the Minimal Resolution Template, and the Stable Atom / Ghost State distinction. Deliverable: Score three real business names from your industry using the four-criterion rubric.

Week 2 — The Cost Variables Sessions: Chapters 5–8. Objective: Calculate α, σ, Φ, trough depth, work-offset ratio, and monthly inference tax for a real named entity. Deliverable: Complete the viability scoring worksheet for a business of your choice.

Week 3 — Differentiation and Compensation Sessions: Chapters 9–11. Objective: Score $\Delta = U \cdot M \cdot T$ for a real business proposition. Understand the Compensation

Theorem and when renaming is preferred over differentiation investment. Deliverable: Run the DuckDuckGo case study analysis independently and compare results to Chapter 11.

Week 4 — Application Sessions: Chapters 12, 18, 20–22. Objective: Complete a full six-step audit on a real named entity and produce a resolution report. Deliverable: Audit report with alternative names scored, trough and viability comparisons, and strategic recommendation.

Structure B: Full-Semester Course (14 Weeks, Undergraduate or Graduate)

Week(s)	Content	Assessment
1–2	Chapters 1–2: Physics of Identity	Reflection paper
3–4	Chapters 3–4: Resolution Template	Name audit #1
5–6	Chapters 5–6: Alignment and Drift	Scoring worksheet
7	Chapter 7–8: Inference Tax and Friction	Cost calculation
8	Midterm examination	Full scoring exam
9–10	Chapters 9–10: Differentiation Theory	Differentiation scoring
11	Chapter 11: DuckDuckGo Case	Case replication
12	Chapters 12–14: Resolution Pressure	Sector pressure mapping
13	Chapters 15–17: Topology / Failure Modes	Failure mode identification
14	Chapters 18–22: Scoring and Application	Final audit project
Finals	Comprehensive audit	Full resolution report

Structure C: Graduate Seminar (8 Weeks, Research Focus)

Weeks 1–2 Foundational theory (Parts I–II) plus the ITT Pre-Veil Mechanics source material. Seminar discussion: is the Minimal Resolution Template a closure theorem or a heuristic? Derive alternative closures for different field structures.

Weeks 3–4 Quantitative framework (Parts III–IV). Problem sets: compute exact compensation requirements for five real businesses; derive the conditions under which differentiation and naming are substitutable vs. non-substitutable.

Weeks 5–6 Resolution Pressure and the Master Equation (Part V). Research project: collect search volume data for three categories, estimate R_p for each, and test whether high-R_p categories show a measurable naming premium in actual search rankings.

Weeks 7–8 Open problems (Chapter 23). Each student selects one open problem and produces a proposal for an empirical research design to address it. Final deliverable: research proposal with formal problem statement, proposed methodology, and expected contribution.

Bibliogr

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This bibliography includes the primary classification system references, the ITT source works that provide the field-theoretic substrate for this framework, and selected commercial and academic works on naming, branding, and semantic classification that informed or contrast with the model developed here.

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